SUPPORTING TOOL

# Architecture of the interactive simulator

A single bilingual (EN/ES) HTML file that runs the normalized formula in real time: every change recomputes the index and all outputs.

**1 · INPUTS**

## 31 sliders

7 at Level 1 (the five components, α and γ) and 24 at Level 2, in four optional blocks: NP, SW, MS and CP can be computed from their sub-components.

→

**2 · ENGINE**

## Normalized formula

```
MP_norm = G^(1−w) · CP^w
```

Weighted geometric mean with $n_{eff} = 3 + \gamma + \alpha$; Proposition 1 live: sign of CP − G.

→

**3 · OUTPUTS**

- Index, level and CP maturity band
- Gauges and radar by component
- Elasticity, marginal return and cost threshold
- Monte Carlo with 20,000 draws
- Comparison charts

**No installation:** it opens in any browser and the whole model runs locally.
diegocabuya.github.io/Cyber_Power_and_Maritime_Power/simulator/simulator.html

# The 31 controls, component by component

| | | | |
|---|---|---|---|
| **MI** | **MI**<br>*Maritime interests* | Direct (weighted sub-index of maritime interests) | **1** |
| **NP** | **$NP = (F \times P \times MDA^{\mu})^{1/(2+\mu)}$**<br>*Naval power* | F · P · MDA · μ | **1 + 4** |
| **SW** | **SW = (PW·IW·SocW)**<br>*Strategic will* | PW · IW · SocW · $w_1$ $w_2$ $w_3$ | **1 + 6** |
| **MS** | **MS = (IC·TE·BC)**<br>*Maritime security* | IC · TE · BC · $a_1$ $a_2$ $a_3$ | **1 + 6** |
| **CP** | **CP = min{1, …}**<br>*Cyber power* | $CS_m$ · $CD_m$ · $RC_m$ · CDip · $\beta_1$ $\beta_2$ $\beta_3$ · δ | **1 + 8** |
| **α γ** | **α · γ**<br>*Parameters* | α: weight of cyber dependence · γ: weight of the security environment | **2** |

**CONTROLS**

**31**

7 at Level 1 + 24 at Level 2

Level-2 values are illustrative and weights are normalized to add up to one.

SIMULATOR · USE

# How it works in practice

1

### Load a scenario

Colombia (hypothetical scenario) or the critical, reference and consolidated profiles of the article.

2

### Adjust the controls

The index, the sensitivity table and the charts recompute instantly.

3

### Read the result

$MP_{norm}$ with its level, the sign of CP − G and the Monte Carlo uncertainty band.

4

### Compare with costs

Use the $c_k/c_{CP}$ threshold of Corollary 1 before prioritizing an investment.

## PRELOADED SCENARIOS

**From the article:** the hypothetical Colombian scenario ($MP_{norm} \approx 0.455$) and the critical ≈ 0.19 · reference ≈ 0.52 · consolidated ≈ 0.85 profiles (Table 6).

**Ten illustrative scenarios:** Brazil, Argentina, US, Canada, Mexico, Chile, Peru, Ecuador, Uruguay and Venezuela, **with a disclaimer**: author-set values, not measurements, not part of the article.

## TEACHING VALUE

Driving any component toward zero collapses the index: limited substitutability stops being a theoretical claim and becomes demonstrable live.